Lydia M. Plante

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Education

Clemson University – BS in Mechanical Engineering, Minor in Music Expected May 2028
GPA: 3.48/4.00 **Honors:** Dean's List (Fall 2024, Spring 2025)
Relevant Coursework: Statics, Dynamics, Materials Science, Engineering Graphics, Differential Equations

Engineering Experience

Structures & Recovery Team Member, Clemson University Rocket Engineering – Clemson, SC September 2025 – Present
- Created designs for structures team airframe force experiments and recovery bay organization under size constraint, progressing airframe thickness reduction and improving structural integration
- Produced manufacturing-ready engineering drawings with GD&T and third-angle projection using Onshape CAD tools
- Fabricated carbon-fiber and fiberglass airframe components, supporting composite layup and assembly processes
- Attended IREC as part of a 14-member team, supporting integration, electronics, and field operations during launch week

Concertmaster, Clemson University Symphony Orchestra – Clemson, SC August 2024 – Present
- Planned and coordinated execution across an 80+ member orchestra, ensuring alignment across sections during rehearsals and performances
- Collaborated with conductor and section leaders to refine musical interpretation and improve performance quality

Work Experience

Supervisor, Cook Mechanical Engineering Lab – Clemson, SC January 2026 – Present
- Instructed students on the safe and effective operation of mechanical engineering lab equipment, including lathes, mills, band saws, welders, and other machining tools
- Digitized certification system by converting paper-based tests to Google Forms and building an automated Google Sheets database, enabling faster tracking, cleaner record-keeping, and seamless integration with the shop's internal certification system.

Repair Technician, Southern String Supply – Mount Pleasant, SC June 2021 – August 2024
- Repaired and refurbished string, brass, and woodwind instruments, restoring functionality and extending lifespan
- Advised customers on instrument selection based on performance needs, improving customer satisfaction

Projects

Four-Bar Linkage Fatigue Tester, Design of Mechanical Systems
- Designed and built a four-bar linkage fatigue testing device to evaluate cyclic loading behavior of materials
- Modeled mechanism kinematics and conducted motion studies in SolidWorks to ensure objectives were met

Rocket-Shaped Container, Engineering Graphics and Machine Design
- Designed a parametric rocket-shaped container in SolidWorks, applying advanced CAD features and design constraints
- Generated fully dimensioned engineering drawings and a 3D printed functional prototype

Egg Beater Reverse Engineering Project, Engineering Graphics and Machine Design
- Reverse engineered a manual egg beater by measuring and modeling individual components in SolidWorks
- Created assembly, motion study, and photorealistic renderings in SolidWorks Visualize to enhance presentation quality and clearly communicate design details

Skills

Software: Onshape, SolidWorks, MATLAB, HTML, CSS, JavaScript, Microsoft Office
Fabrication & Manufacturing: Machine Shop (Certified), Welding (Certified), Composite Manufacturing
Engineering Skills: Engineering Drawings, CAD Modeling, Prototyping, Data Analysis
Languages: Spanish (Proficient)